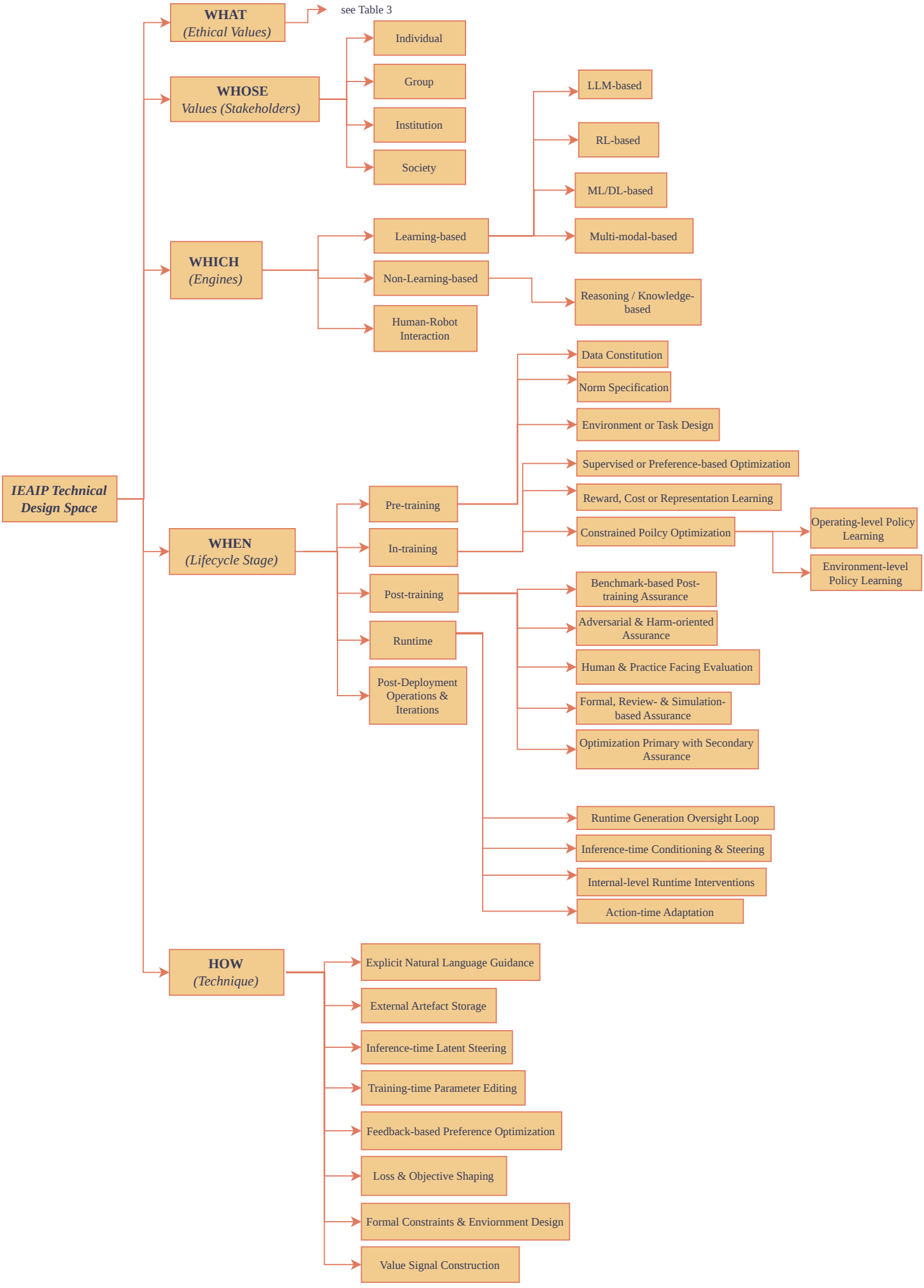

IEAIP Technical Design Space
WHAT (Ethical Values)
see Table 3
WHOSE Values (Stakeholders)
Individual
Group
Institution
Society
WHICH (Engines)
Learning-based
Non-Learning-based
Human-Robot Interaction
LLM-based
RL-based
ML/DL-based
Multi-modal-based
Reasoning / Knowledge-based
WHEN (Lifecycle Stage)
Pre-training
In-training
Post-training
Runtime
Post-Deployment Operations & Iterations
Data Constitution
Norm Specification
Environment or Task Design
Supervised or Preference-based Optimization
Reward, Cost or Representation Learning
Constrained Poilcy Optimization
Operating-level Policy Learning
Environment-level Policy Learning
Benchmark-based Post-training Assurance
Adversarial & Harm-oriented Assurance
Human & Practice Facing Evaluation
Formal, Review- & Simulation-based Assurance
Optimization Primary with Secondary Assurance
Runtime Generation Oversight Loop
Inference-time Conditioning & Steering
Internal-level Runtime Interventions
Action-time Adaptation
HOW (Technique)
Explicit Natural Language Guidance
External Artefact Storage
Inference-time Latent Steering
Training-time Parameter Editing
Feedback-based Preference Optimization
Loss & Objective Shaping
Formal Constraints & Enviornment Design
Value Signal Construction